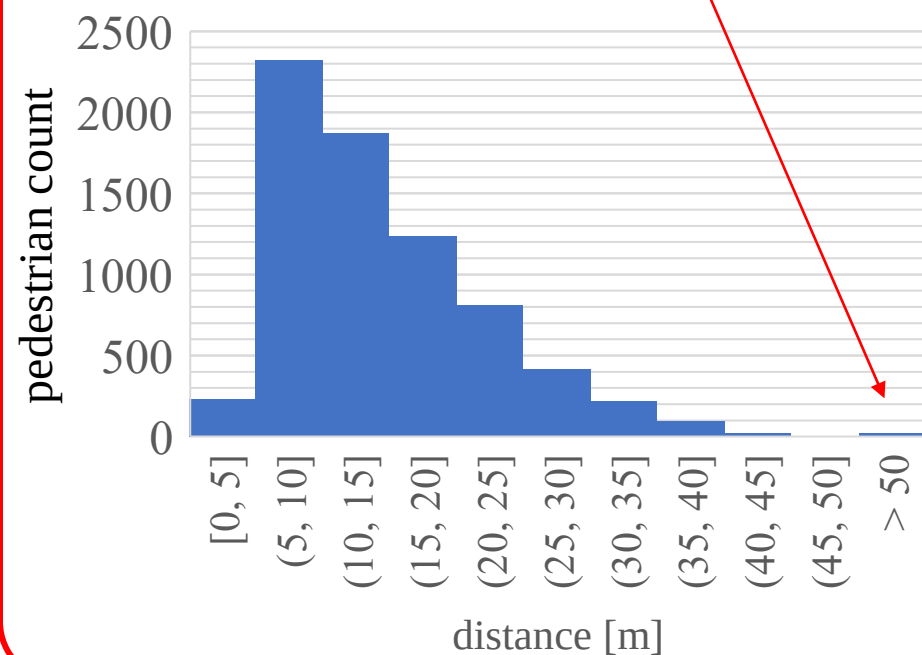


Road Scene Pedestrian Relocation for Data Augmentation

Chen Shi

Far Pedestrian Detection Problem



DANGEROUS!!

- 100km/h $\approx$ 50m in 2s

WHY?

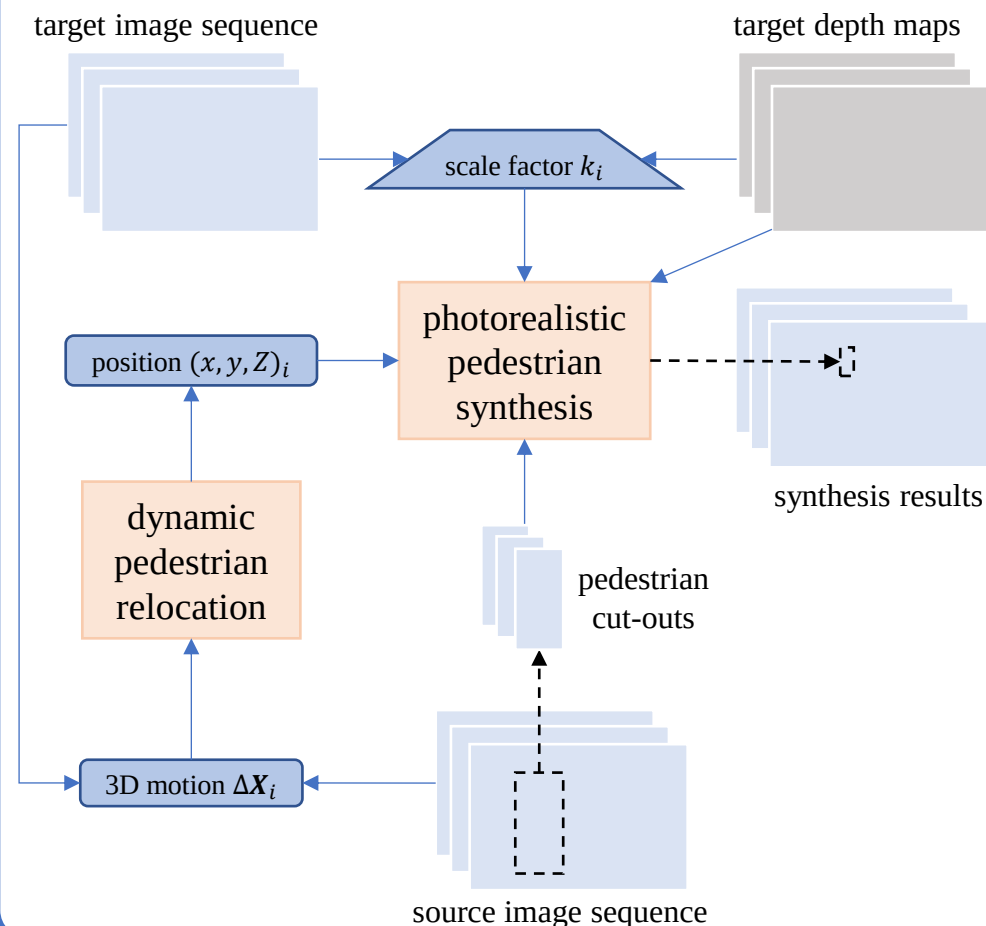
- Low resolution
- **UNANNOTATED FAR PEDESTRIANS**

HOW?

- Data augmentation
- Add annotated far pedestrians

Pedestrian Relocation

- Cut out annotated pedestrians
- Synthesize into new scenes with correct scale and occlusion
- Sequence-to-sequence



One-frame Examples



Experimental Result

Mask R-CNN shows +1.117% and +0.26% AP improvement in “far pedestrian” (≤ 50 pixels tall) detection and instance segmentation, respectively.

Multi-frame Example

